ROBOTICS · CHART 1

ESP32 car + ESP32-C3 gateway

MarkerPilot

A robot car that drives itself, guided by one camera on the ceiling

Camera sees → laptop thinks → radio → car drives

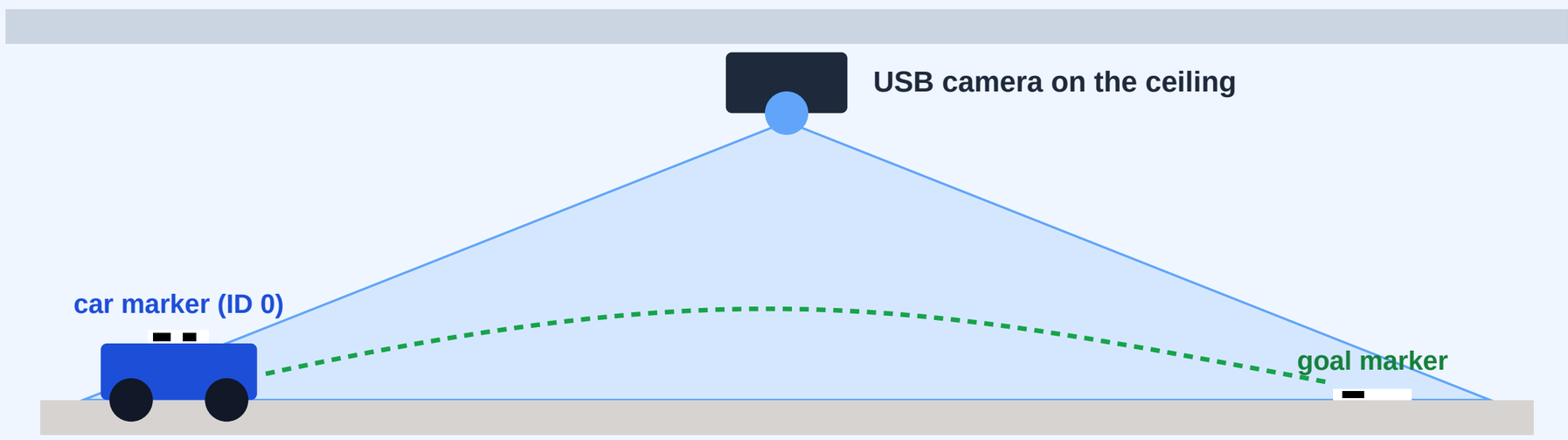
1 The problem

- Warehouse robots need costly sensors (LiDAR, encoders).
- Wheel slip makes their position drift.
- Too expensive for schools and small shops.

2 Our idea

- One cheap **ceiling camera** sees the whole floor.
- A printed **ArUco marker** shows where the car is.
- Click anywhere - the car drives there.

3 How the car is seen

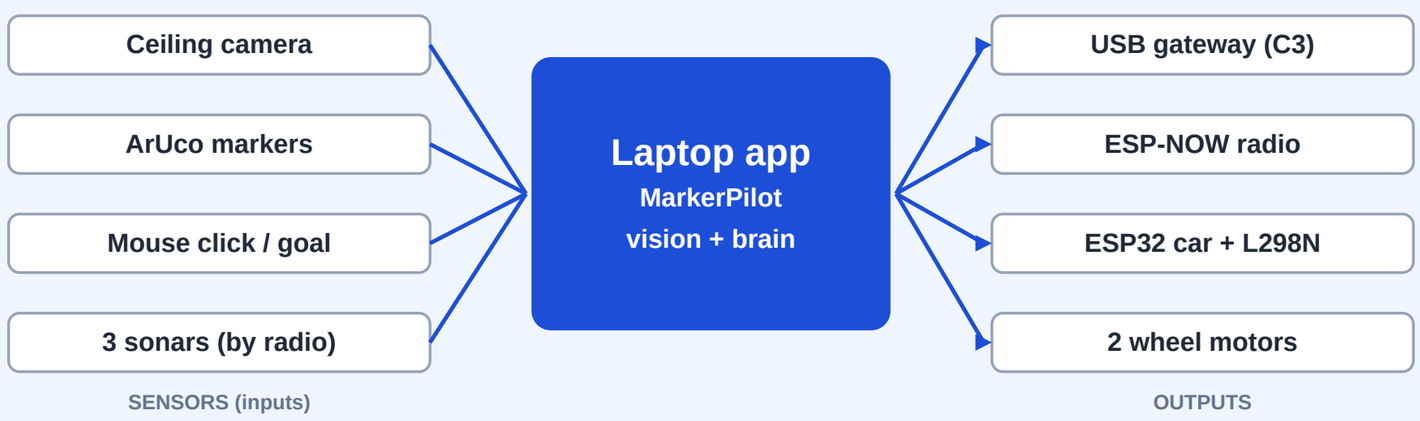


the car drives itself to the goal or to any point clicked on the screen

4 Parts we used

Part	Job	Rs.
ESP32-WROOM-32	car: motors + sensors	420
ESP32-C3 SuperMini	USB radio gateway	300
L298N	motor driver	180
2 geared DC motors	wheels	160
3 x HC-SR04	obstacles L / F / R	270
1080p USB webcam	ceiling camera	1,200
2S Li-ion battery	power	360
Chassis, markers, wires		370
Total		3,260

5 What makes it smart

- **Kalman filter**: smooth position, predicts 0.08 s ahead.
- **Two brains**: classic controller and an **RL agent** that learned by practice.
- **Safety**: sonars steer round obstacles; the car refuses to hit anything close.
- **Simulator** with real ArUco pictures.

6 How the parts connect